SOLUTIONS: DISCRETE EVENT SYSTEMS
MASTER IN CONTROL

1. Exercise

a) The automata G_A and G_B, modeling the 2 queues, are shown in Fig. 1.1. [2]

b) Sometimes events of type x occur. These exceptions can be taken into account as shown in Fig. 1.2. [2]

c) Parallel composition $\tilde{G}$ is shown in Fig. 1.3. [4]

d) The labelling automaton is shown in Fig. 1.4 [2]

e) The parallel composition $\tilde{G}||G_L$ is shown in Fig. 1.5. [3]

f) The non-deterministic automaton G_N is shown in Fig. 1.6. [1]

g) The diagnoser automaton is computed by constructing Obs(G_N) and is shown in Fig. 1.7. [6]

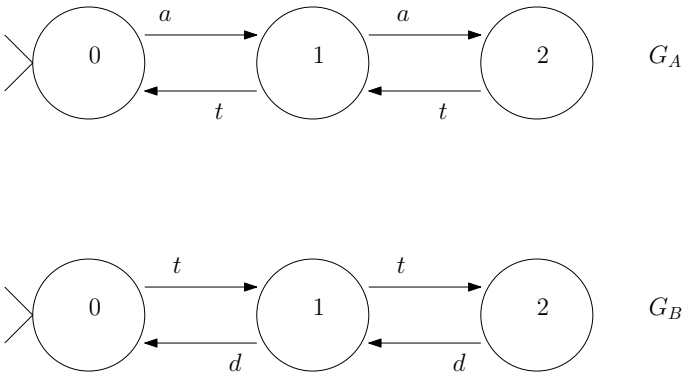


Figure 1.1 Automata modelling QA and QB

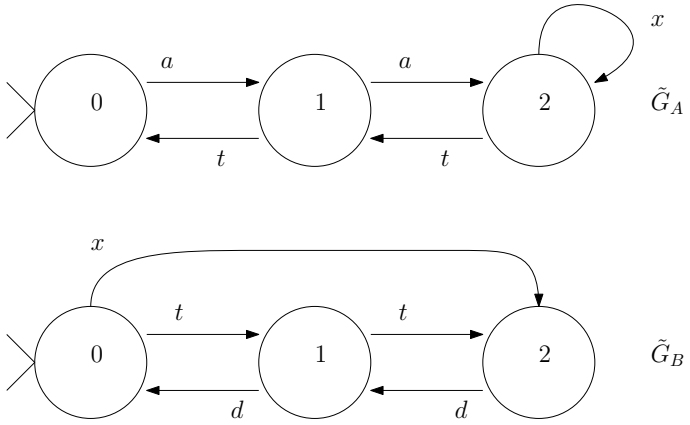


Figure 1.2 Automata modelling QA and QB with exceptions

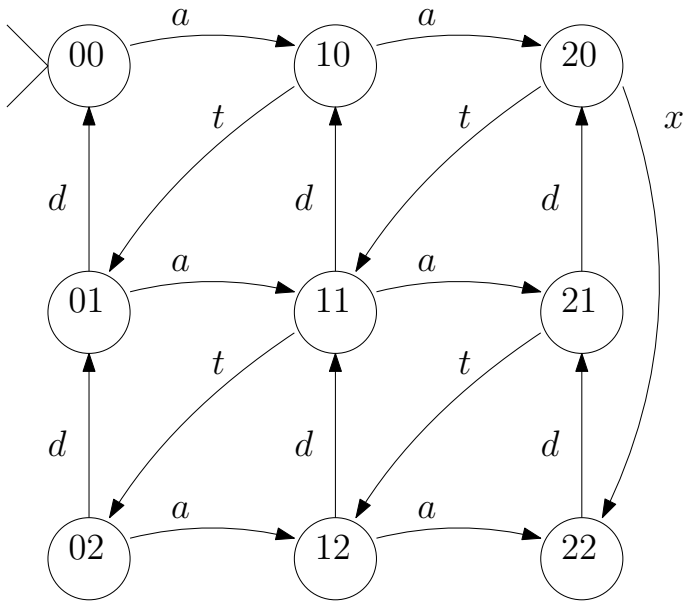


Figure 1.3 Automaton $\tilde{G}$

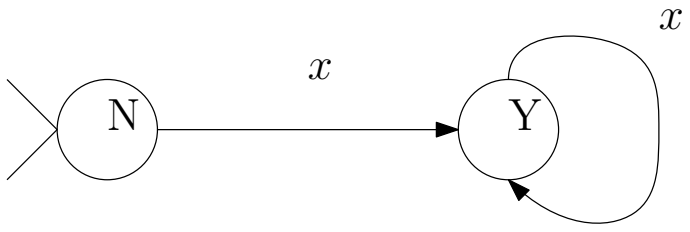


Figure 1.4 The labelling automaton G_L

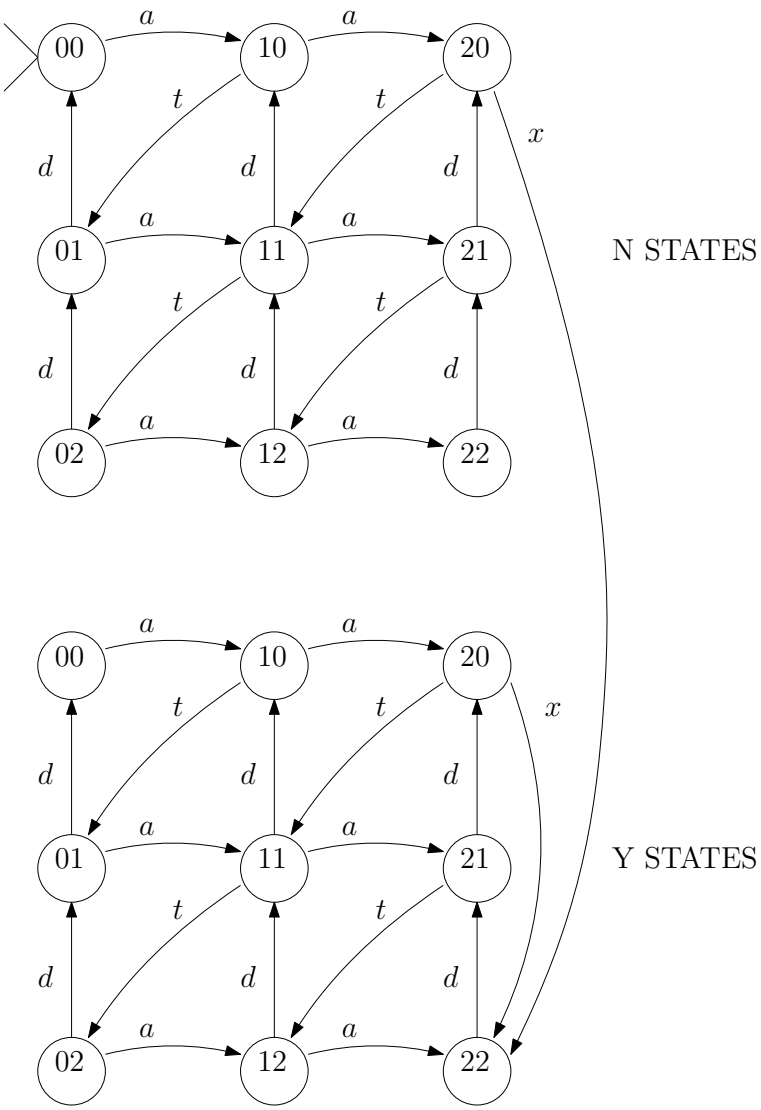


Figure 1.5 The parallel composition $\tilde{G}||G_L$

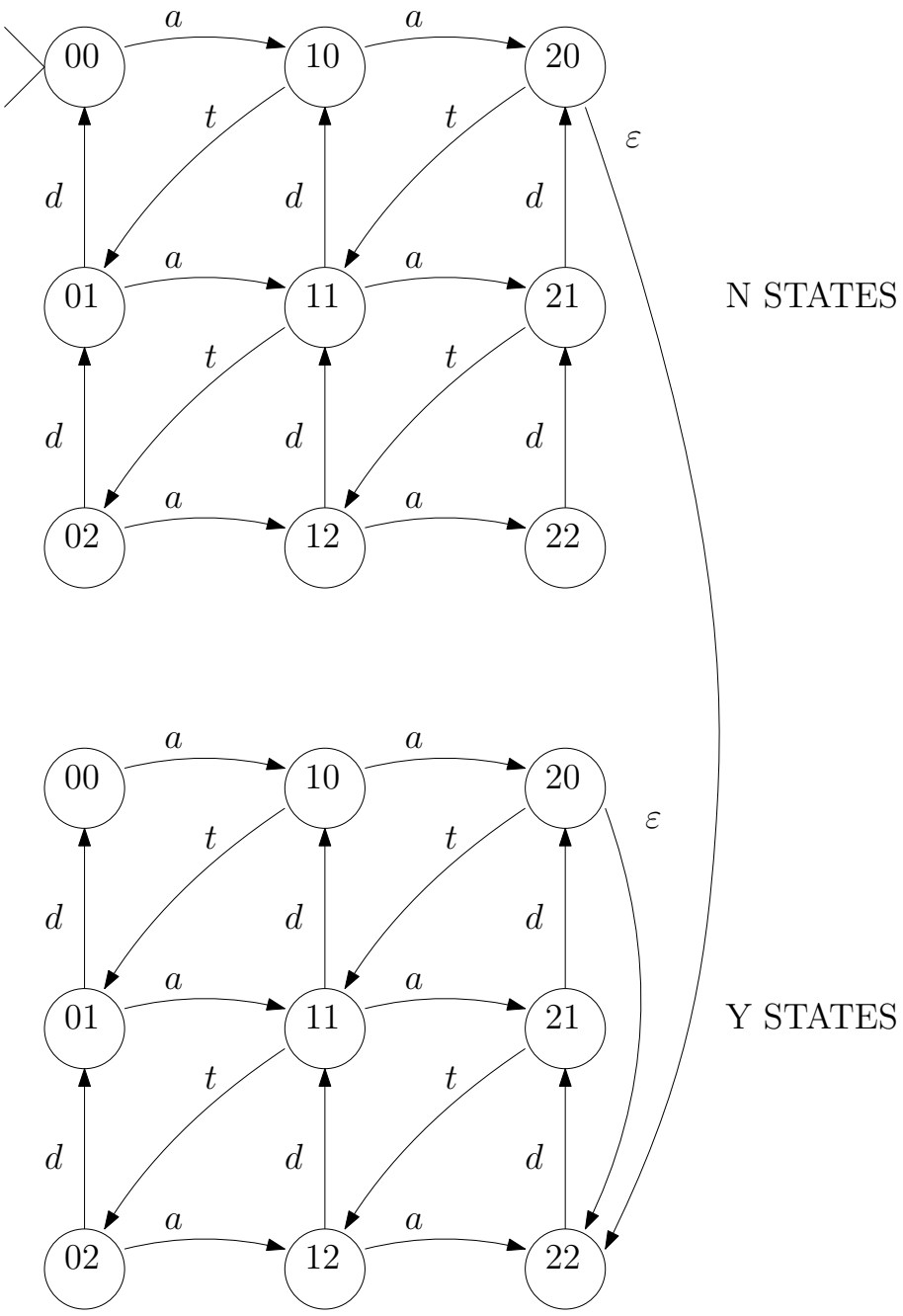


Figure 1.6 The non deterministic automaton G_N

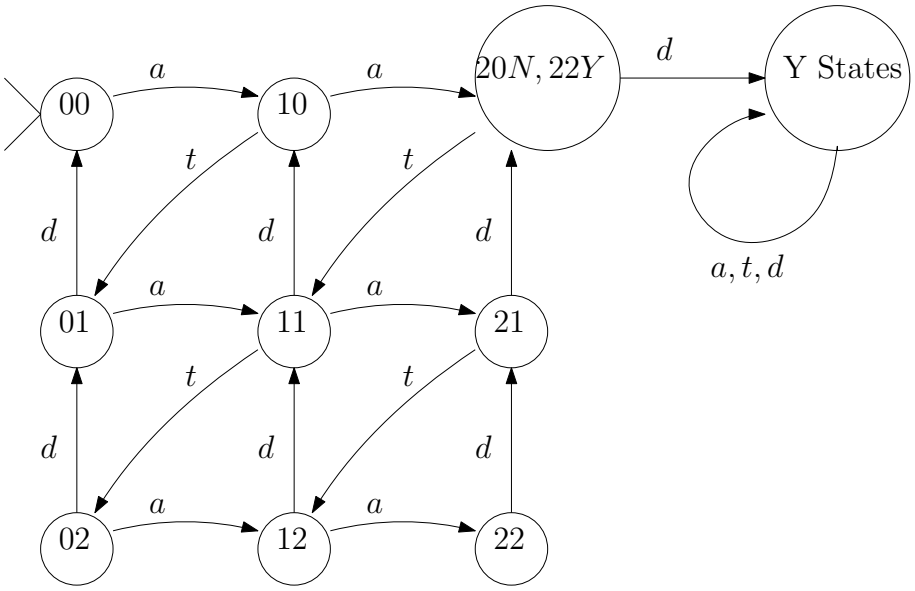


Figure 1.7 The diagnoser automaton Obs(G_N)

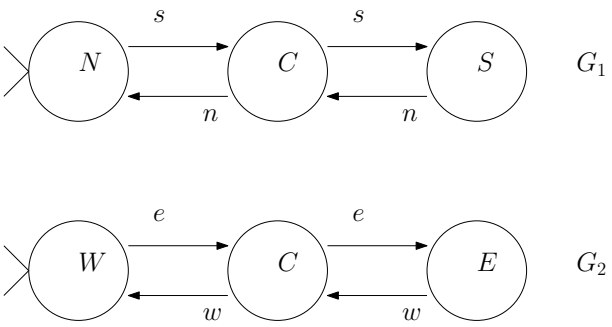


Figure 2.1 Transition diagrams of G_1 and G_2

2. Exercise

a) Each automaton has 3 states, one per working cell. The corresponding transition diagrams are shown in Fig. 2.1. [3]

b) The parallel composition of G_1 and G_2 is given in Fig. 2.2. [3]

c) There is only one state in $G_1||G_2$ where both robots are in the same cell. This is state *CC*. Hence, the specification is an "illegal state" specification and can simply be obtained from the automaton $G_1||G_2$ by removing the forbidden state and all incidents transitions (see Fig. 2.3). [4]

d) To verify controllability of the specification we compute the parallel composition $G_1||G_2||G_S$ and verify the difference between the original set of enabled events and the events enabled after parallel composition, in each of the states. The specification is uncontrollable as event *e* is disabled in (CW,CW), and similarly event *n* is disabled in (SC,SC) (see Fig. 2.4). [3]

e) The supervisor implementing the maximal controllable sublanguage is shown in Fig. 2.5. [4]

f) When NW is the only terminal state, the previous supervisor already achieves a trim automaton. Hence, the non-blocking supervisor is the same as in the

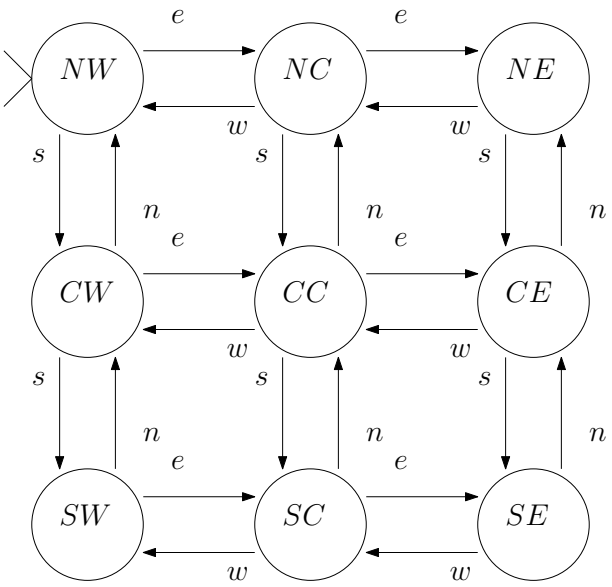


Figure 2.2 Transition diagrams of $G_1||G_2$

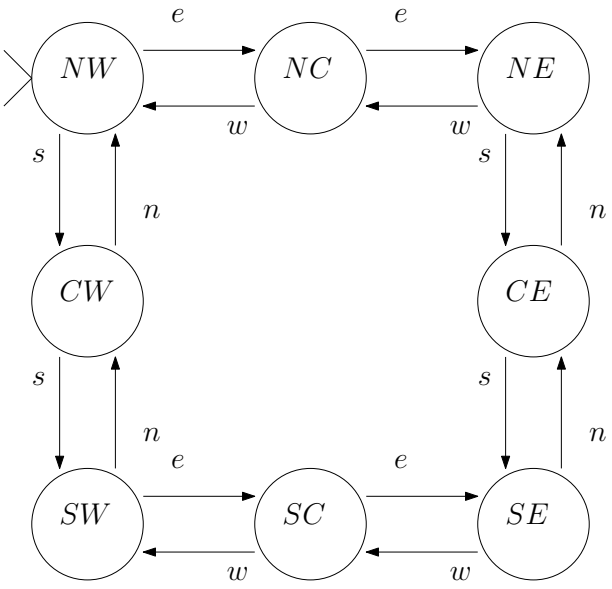


Figure 2.3 Illegal state specification automaton G_S

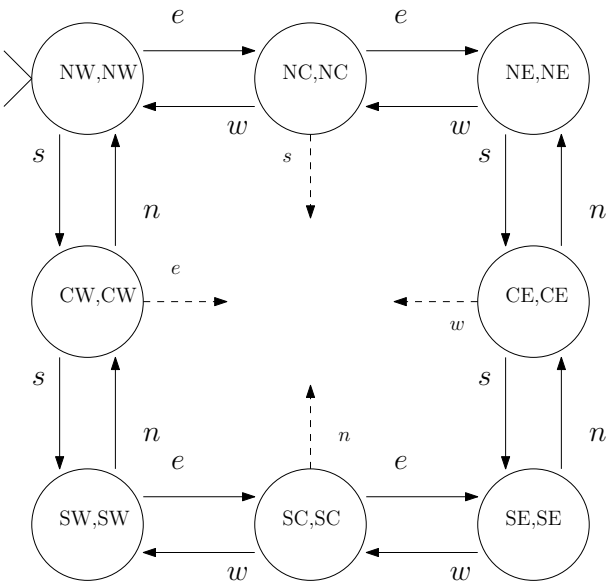


Figure 2.4 Parallel composition $G_1||G_2||G_S$

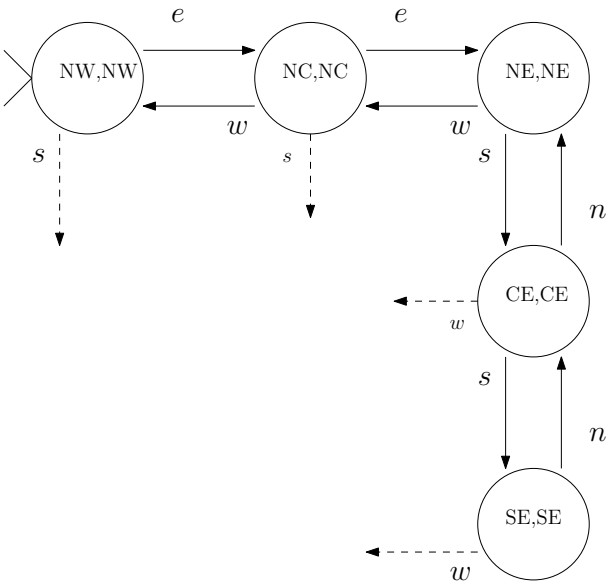


Figure 2.5 Supervisor implementing the maximal controllable sublanguage

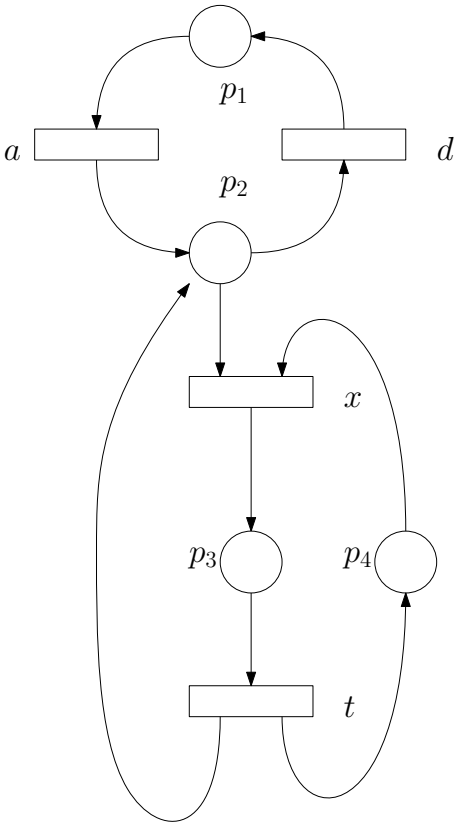


Figure 3.1 The Petri Net N_1

previous question. [3]

3. Exercise

a) The Petri Net N_1, modelling the Museum, is shown in Fig. 3.1. In particular, there are four transitions $T = \{a, d, x, t\}$ modelling each the corresponding event. There are 4 places in the network, p_1 represents the residual museum capacity, and is initialized at C_M. Place p_2 represents the number of visitors inside the museum, but not in the exhibition, and is initialized with 0 tokens. Place p_3 represents the number of visitors in the exhibition (initialized at 0), and p_4 the residual capacity of the exhibition and is initialized with C_E tokens. (4 marks for correct Petri Net and 2 marks for correct initialization) [6]

b) The incidence matrix C, listing transitions in the order a, d, x, t, is given as:

$$C = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

[2]

c) There are 2 minimal semiflows in the network, the vectors $[1,1,1,0]$ and $[0,0,1,1]$. The first represents the conservation of people inside the museum (including the exhibition) and residual museum capacity. The second represents conservation of people inside the exhibition and exhibition residual capacity. [3]

d) The place p_5 is added so as to increase by 1 token each time a new arrival occurs. See Fig. 3.2. [2]

e) An extra transition can be modeled to increase capacity by 1 as in Fig. 3.3.

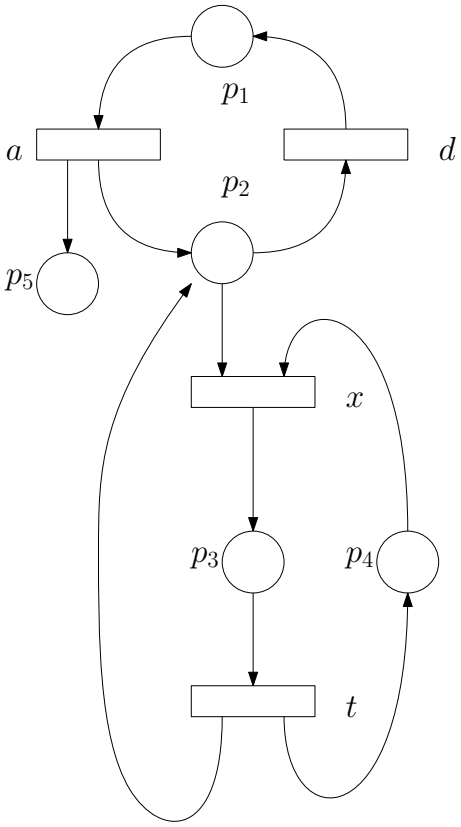


Figure 3.2 Petri Net with ticket counter

Notice that its precondition has a weight of 1000. [2]

f) The Petri Net N_1 is structurally bounded as $[1,1,1,0]+[0,0,1,1]=[1,1,2,1]$ is a P-semiflow with support equal to the set of all places. On the other hand, the C matrix for N_2 is given as:

$$C = \begin{bmatrix} -1 & 1 & 0 & 0 & 1 \\ 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 0 & -1000 \end{bmatrix}.$$

We claim that N_2 is not structurally bounded. Notice that the vector $v=[1000,1000,0,0,1]'$ is a T-increasing vector. Indeed: $Cv=[1,0,0,0,0]'$. Hence no P-vector w of full support exists such that $wC \leq 0$, since $0 \geq wCv = w_1 > 0$ which is absurd. This corresponds to the intuition that an unlimited number of tickets can be sold and also capacity can grow unbounded. [5]

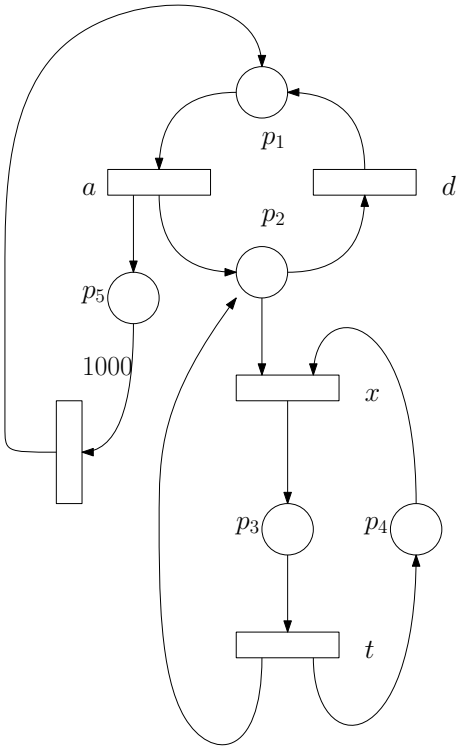


Figure 3.3 Petri Net N_2

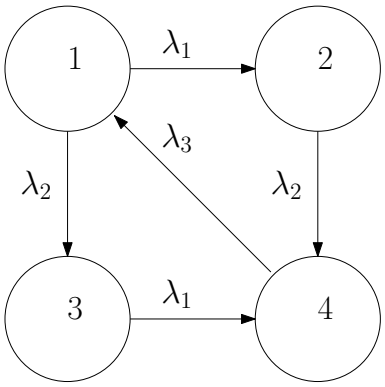


Figure 4.1 Transition diagram of Markov Chain

4. Exercise

a) To model the machine we use a Markov chain with 4 states. In state 1 both arrival processes are acting, and transition to state 2 or 3 may occur depending on which piece arrives first. Then, transition to state 4 occurs from both states with the rate corresponding to the missing piece. Finally from state 4 the cycle is completed with a transition towards state 1 at rate λ_3. A diagram is shown in Fig. 4.1. The Kolmogorov equations describing the state evolution are given as:

$$\dot{\pi} = \pi Q$$

$$Q = \begin{bmatrix} -\lambda_1 - \lambda_2 & \lambda_1 & \lambda_2 & 0 \\ 0 & -\lambda_2 & 0 & \lambda_2 \\ 0 & 0 & -\lambda_1 & \lambda_1 \\ \lambda_3 & 0 & 0 & -\lambda_3 \end{bmatrix}.$$

[4]

b) The Markov chain is ergodic as its transition diagram is strongly connected and therefore has a unique absorbing strongly connected component. [2]

c) To find steady-state probability distribution we need to solve the following system of linear equations:

$$\pi_\infty Q = 0 \qquad \pi_\infty \mathbf{1} = 1,$$

where $\mathbf{1}$ is a column vector of ones (of dimension 4). Accordingly we see that:

$$\pi_{1\infty} = \frac{\lambda_1\lambda_2\lambda_3}{\lambda_1\lambda_2\lambda_3 + \lambda_1^2\lambda_3 + \lambda_2^2\lambda_3 + \lambda_1^2\lambda_2 + \lambda_1\lambda_2^2},$$

$$\pi_{2\infty} = \frac{\lambda_1}{\lambda_2}\pi_{1\infty}$$

$$\pi_{3\infty} = \frac{\lambda_2}{\lambda_1}\pi_{1\infty}$$

$$\pi_{4\infty} = \frac{\lambda_1+\lambda_2}{\lambda_3}\pi_{1\infty}$$

The average fraction of time that arrival 1 is interrupted is given by $\pi_{2\infty} + \pi_{4\infty}$ as 2 and 4 are the states where arrivals are not happening. [4]

d) The final part of the cycle has average duration $1/\lambda_3$. Hence to find the average cycle duration we only need to add (after removing the state transition from 4 to 1) the average absorption time from initial state 1 to final state 4. We wish to compute $\int_0^{+\infty} t\dot{\pi}_4(t)dt$, and we do this in the frequency domain:

$$\mathscr{L}[\dot{\pi}_4] = [1,0,0](sI - \tilde{Q})^{-1}\begin{bmatrix} 0 \\ \lambda_2 \\ \lambda_1 \end{bmatrix} = \frac{\lambda_1\lambda_2(2s+\lambda_1+\lambda_2)}{(s+\lambda_1)(s+\lambda_2)(s+\lambda_1+\lambda_2)}$$

where $\tilde{Q}$ is the principal minor of order 3 of Q. Finally the average absorption time is given as

$$E[\tau] = -\frac{d}{ds}\mathscr{L}[\dot{\pi}_4]\Big|_{s=0} = \frac{\lambda_1^2+\lambda_1\lambda_2+\lambda_2^2}{\lambda_1\lambda_2(\lambda_1+\lambda_2)}$$

and the average cycle duration is simply obtained by adding $1/\lambda_3$. [4]

e) A state 5 can be added to the Markov Chain, corresponding to the situation where there is an arrival of piece 1 during the assembling operation (thus filling up the buffer). Transition out of state 5 is still at a rate λ_3 and will then lead to state 2 where piece 1 is already available and only piece 2 needs waiting for. See Fig. 4.2. [3]

f) The fraction of time can now be computed as $\pi_{2\infty} + \pi_{5\infty}$ where π_∞ is solution to: $\pi_\infty Q = 0$, $\pi_\infty \mathbf{1} = 1$ for Q modified as:

$$Q = \begin{bmatrix} -\lambda_1-\lambda_2 & \lambda_1 & \lambda_2 & 0 & 0 \\ 0 & -\lambda_2 & 0 & \lambda_2 & 0 \\ 0 & 0 & -\lambda_1 & \lambda_1 & 0 \\ \lambda_3 & 0 & 0 & -\lambda_3-\lambda_1 & \lambda_1 \\ 0 & \lambda_3 & 0 & 0 & -\lambda_3 \end{bmatrix}.$$

[3]

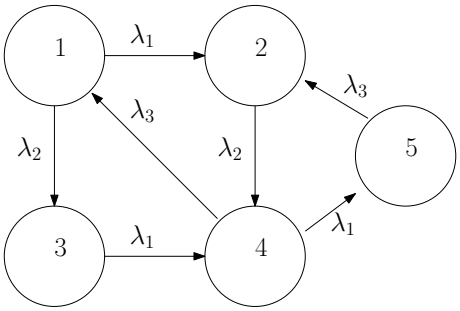


Figure 4.2 Modified Markov Chain